RISHU KUMAR SINGH

Kolkata, India | +91 9801069471 | rishukumarsingh1819@gmail.com
LinkedIn: linkedin.com/in/rishu-kumar-singh-3aa159365 | GitHub: github.com/rishukumarsingh123 | Portfolio: rishukumarsingh123.github.io/MyPortfolio

PROFESSIONAL SUMMARY

B.Tech Computer Science (AI) student with an 8.70 CGPA, strong foundation in Data Structures & Algorithms, and practical experience in full-stack web and IoT development. Solved 400+ problems across platforms like LeetCode, GeeksforGeeks, and HackerRank. Seeking software engineering opportunities to apply problem-solving skills and build scalable web applications.

EDUCATION

Institute of Engineering and Management (NewTown), Kolkata — **2024 – 2028**
Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence) — **CGPA: 8.70 / 10**

St. Xavier's High School, Patna, India — **2021 – 2023**
Higher Secondary (Class XII) — **Percentage: 82.2%**

High School Satjora, Saran, India — **2020 – 2021**
Secondary School (Class X) — **Percentage: 85.2%**

TECHNICAL SKILLS

Languages: C++, Java, Python, C, JavaScript, SQL
Frontend: HTML5, CSS3, JavaScript, React.js, Tailwind CSS
Backend: Node.js, Express.js, REST APIs
Databases: MongoDB, MongoDB Atlas, MySQL
Core Fundamentals: Data Structures and Algorithms (DSA), Object-Oriented Programming (OOP)
Tools & Platforms: Git, GitHub, VS Code, Postman, Vercel

PROJECTS

EcoGrid — Smart Waste Management and Monitoring System | *ESP32, C++, Node.js, Express.js, MongoDB, React.js*

- Developed an IoT waste monitoring system utilizing ESP32 to monitor bin fill capacity and gas levels in real time.
- Integrated ultrasonic sensors for distance-based fill calculation and MQ135 sensor for hazardous odor/gas detection.
- Built REST APIs with Node.js and Express.js to process incoming sensor telemetry and persist records in MongoDB.
- Designed a responsive React.js dashboard providing live status monitoring, location tracking, and collection workflow.
- Implemented ESP32 firmware in C++ for sensor reading, fill-level calculations, and Wi-Fi data transmission.

Weather Application | *HTML5, CSS3, JavaScript, OpenWeather REST API*

- Developed a responsive client-side weather forecasting application integrating third-party REST APIs.
- Implemented asynchronous API retrieval and dynamic DOM updates to display real-time temperature, wind, and humidity.
- Created an intuitive user interface with dark-mode support and cross-device responsive styling.

CODING & PROBLEM SOLVING

Codolio Profile: codolio.com/profile/rishu1819

- Solved **400+ problems** focusing on Data Structures and Algorithms across LeetCode, GeeksforGeeks, and HackerRank.
- Practiced core DSA topics: Arrays, Strings, Two Pointers, Sliding Window, Linked Lists, Stacks, Queues, Trees, Graphs, and Dynamic Programming.
- Regularly solve algorithmic problems in C++ with an emphasis on optimal time and space complexity.

AREAS OF INTEREST

Software Engineering | Full-Stack Development | Backend Development | Data Structures and Algorithms